I've Got a Categorical Variable?!



Now What?

Andrew Zieffler
Department of Educational Psychology

Scales of Measurement

Classification system describing the nature of information within the values assigned to variables. (Stevens, 1946)

Categorical Data is Here

Scale	Property	Operations	Examples
Nominal	Class membership	= and ≠	College major, Sex, Political affiliation
Ordinal	Comparison	< and >	Likert data, Rankings, Scoville scale
Interval	Difference	+ and −	GRE scores, Temperature (F)
Ratio	Magnitude	× and ÷	Income, Class size, Years of experience

Categorical variables are at the nominal scale of measurement, and although, in practice, we assign numbers to represent the levels of the categorical variable, those numbers do not carry any more information than **group membership**.

Dichotomous and Polytomous Variables

When a categorical variable has two levels we refer to it as **dichotomous** or **binary**. If it has more than two levels, it is **polychotomous** or **polytomous**.

Dichotomous

STEM Major?
STEM
Non-STEM
Non-STEM
STEM
STEM
STEM
Non-STEM

Polytomous

Major
Kinesiology
Special Education
Special Education
Child Psychology
Kinesiology
Kinesiology
Child Psychology

Contingency Tables

A contingency table is simply a table that lists each level of the categorical variable and those level's counts / percentages.

Data

Major
Kinesiology
Special Education
Special Education
Child Psychology
Kinesiology
Kinesiology
Child Psychology

Contingency Table

Major	Count
Kinesiology	3
Special Education	2
Child Psychology	2

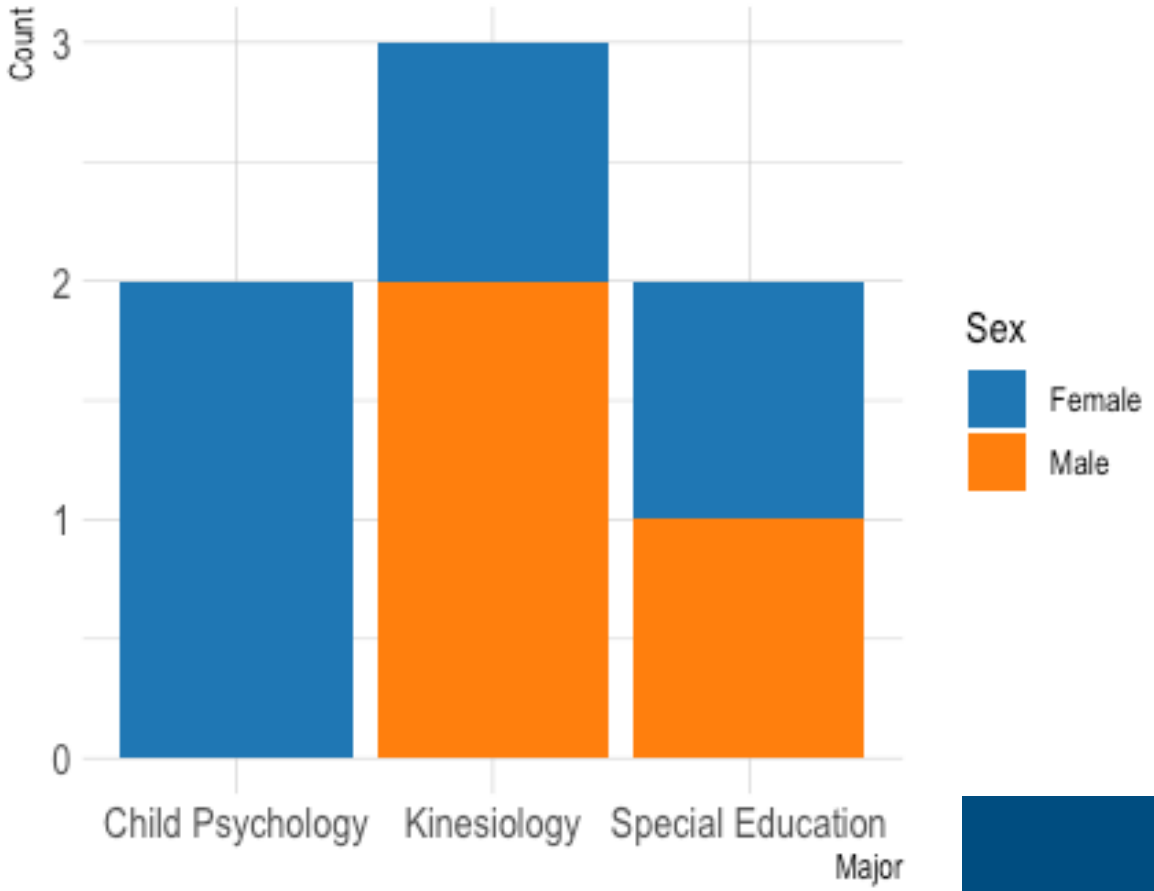
Contingency tables can also be used to show cross-classifications of two (or more) variables.

Data

Major	Sex
Kinesiology	Female
Special Education	Female
Special Education	Male
Child Psychology	Female
Kinesiology	Male
Kinesiology	Male
Child Psychology	Female

Contingency Table

	Sex		
Major	**Female**	**Male**	**Total**
Kinesiology	1	2	***3***
Special Education	1	1	***2***
Child Psychology	2	0	***2***
Total	***4***	***3***	***7***

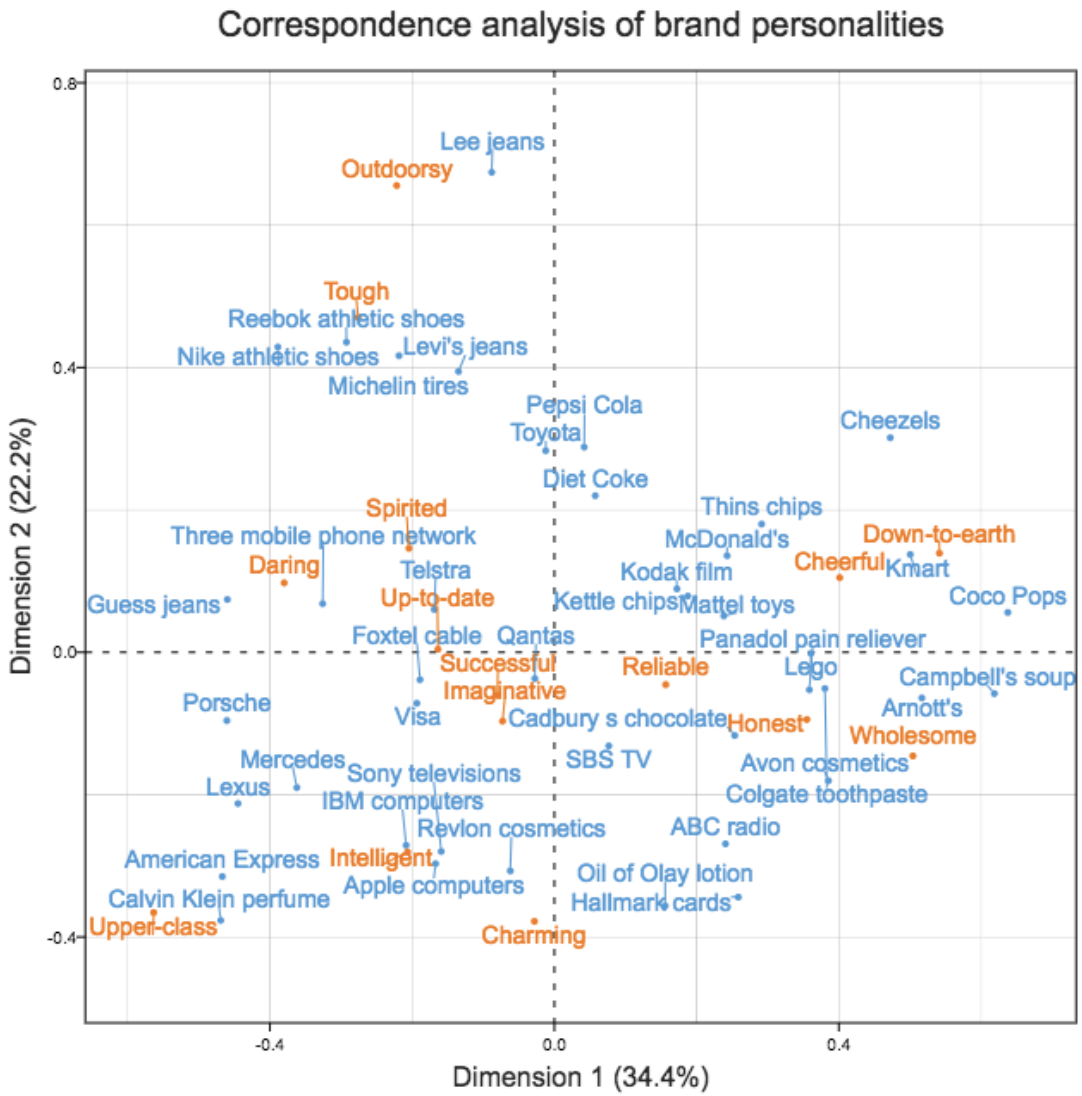


Bar charts are just graphical summaries of the information in a contingency table.

Major	Sex		Total
	Female	Male	
Kinesiology	1	2	***3***
Special Education	1	1	***2***
Child Psychology	2	0	***2***
Total	***4***	***3***	***7***

Many methods of analyzing categorical data are based off of contingency tables:

- Methods of association
  - Chi-squared () statistics and tests
  - Phi coefficient
  - Tetrachoric correlation
  - Cramer's V
  - Goodman & Kruskal's lambda
  - Goodman & Kruskal's gamma
  - Kendall's tau
- Log-linear modeling
- Correspondence analysis



Challenge #1: Many common statistical methods require quantitative variables

To alleviate this problem, we typically re-code (or treat) categorical variables so that they are quantitative. (Remember: *The numbers only denote group membership*.)

ID	Major	Recoded 1	Recoded 2	Recoded 3
1	Kinesiology	1	1	–1
2	Special Education	2	100	0
3	Special Education	2	100	0
4	Child Psychology	3	3000	1
5	Kinesiology	1	1	–1
6	Kinesiology	1	1	–1
7	Child Psychology	3	3000	1

Dummy/Reference Coding

One common method for coding categorical variables is **dummy coding** (aka, **reference coding**).

- Dummy coding only uses the values 0 and 1
- Typically 1 denotes membership in a particular level and 0 indicates not a member of that level

ID	Graduation Status	Graduated
1	Graduated	1
2	Did not graduate	0
3	Graduated	1
4	Graduated	1
5	Graduated	1
6	Did not graduate	0
7	Graduated	1
8	Did not graduate	0
9	Graduated	1
10	Graduated	1

Dummy coding has several useful advantages. For example, *the mean of a dummy coded variable is the proportion of cases coded as 1.*

ID	Graduation Status	Graduated
1	Graduated	1
2	Did not graduate	0
3	Graduated	1
4	Graduated	1
5	Graduated	1
6	Did not graduate	0
7	Graduated	1
8	Did not graduate	0
9	Graduated	1
10	Graduated	1

$$\text{Mean} = \frac{1+0+1+1+1+0+1+0+1+1}{10} = \frac{7}{10} = 0.7$$

The proportion of students who graduated is 0.7.

With polytomous variables we need to use more than one dummy variable to code all of the categories.

To distinctly code all of the categories we need to create a dummy variable for *all categories except one*.

ID	Major	Kinesiology	Spec_Ed
1	Kinesiology	1	0
2	Special Education	0	1
3	Special Education	0	1
4	Child Psychology	0	0
5	Kinesiology	1	0
6	Kinesiology	1	0
7	Child Psychology	0	0

- *Kinesiology majors:* `Kinesiology = 1` and `Spec_Ed = 0`
- *Special Education majors:* `Kinesiology = 0` and `Spec_Ed = 1`
- *Child Psychology majors:* `Kinesiology = 0` and `Spec_Ed = 0`

Because there are multiple dummy variables, we compute multiple means.

$$\text{Kinesiology} = \frac{1+0+0+0+1+1+0)}{7} = \frac{3}{7} = 0.429$$

$$\text{Special Education} = \frac{0+1+1+0+0+0+0)}{7} = \frac{2}{7} = 0.286$$

- The proportion of Kinesiology majors is 0.429.
- The proportion of Special Education majors is 0.286.

$$\text{Child Psychology} = 1 - 0.429 - 0.286 = 0285$$

- The proportion of Child Psychology majors is 0.285.

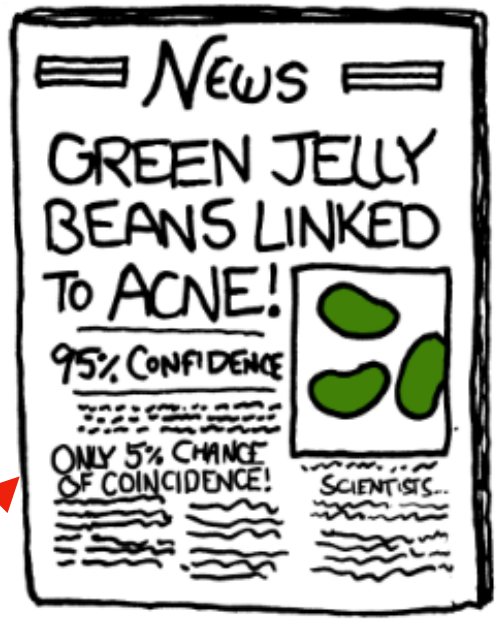
Challenge #2: Categorical variables may have many levels

Imagine if we had a variable `state` that we wanted to analyze. If we were to code this into a set of dummy variables, we would need to create 49 dummy variables! (Or 51 if we include Puerto Rico and Washington, DC.)

ID	State	MN	NY	CA
1	Minnesota	1	0	0
2	New York	0	1	0
3	California	0	0	1
4	Iowa	0	0	0
5	North Dakota	0	0	0
6	Texas	0	0	0
7	Oregon	0	0	0
⋮	⋮	⋮	⋮	⋮

Need to add 46 more dummy variables…





If we use a variable with many levels in an analysis (say we want to see if there are differences in ACT scores across states), we will need to adjust our *p*-values to **account for the high number of comparisons** (e.g., Bonferroni adjustment).

Potential Solution: Collapse the variable into fewer categories by combining several categories into a single category.

For our state example we might collapse states into regions. This reduces the number of levels from 50 to 4 or 5 (depending how many regions we envision).

ID	State	Region
1	Minnesota	Midwest
2	New York	East
3	California	West
4	Iowa	Midwest
5	North Dakota	Midwest
6	Texas	South
7	Oregon	West
⋮	⋮	⋮

Challenge #3: One or more categories are very rare

If one or more categories have very few cases relative to others, they will offer little to no information in the analysis (too little variation). In some cases, models may fail to converge.



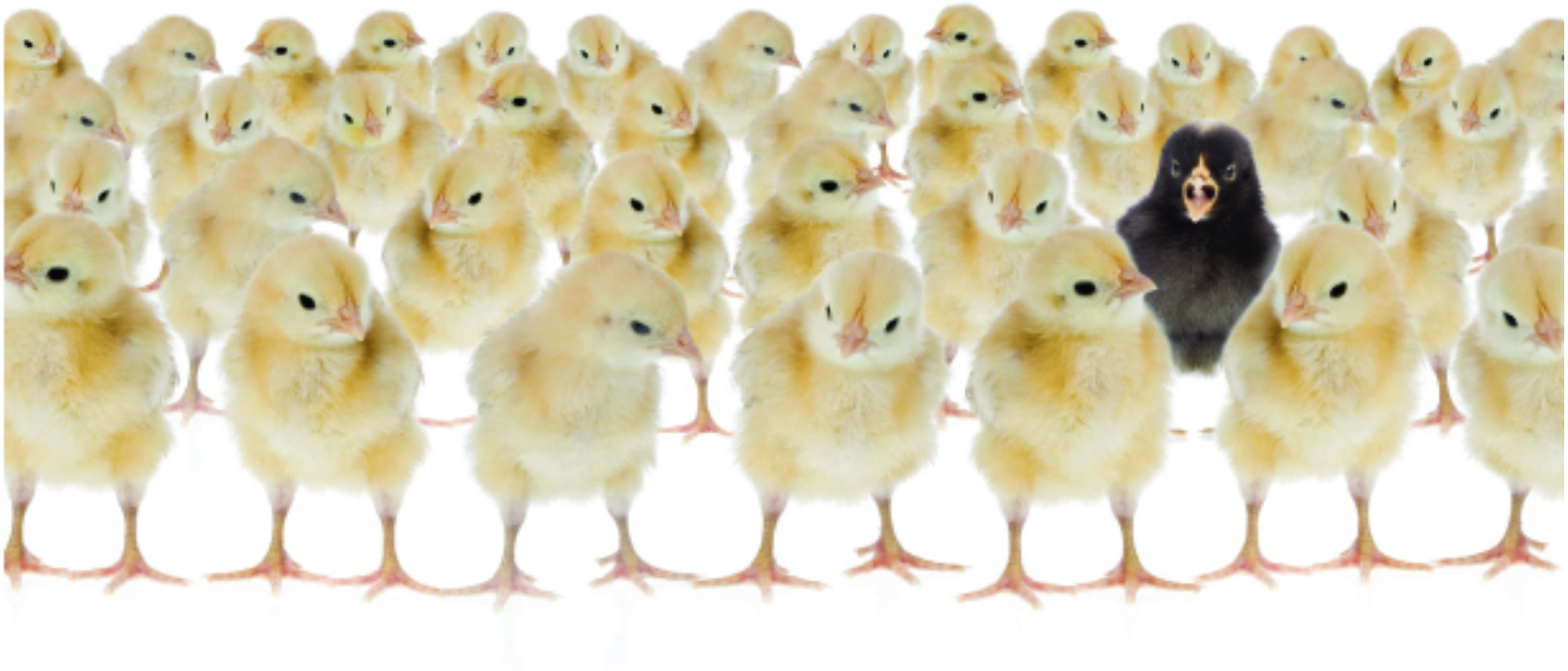
Potential Solution: Try to collapse these categories into other categories.

Survey responses might allow respondents to write-in information. Below Respondent #2 chose to write in her/his/their race/ethnicity. This could be (depending on the RQ) collapsed into an "Other" category along with other write-in responses that cannot be categorized.

ID	Self Identified Race/Ethnicity	Collapsed Race
1	Hispanic	Hispanic
2	African Cuban	Other
3	White	White
4	African American	African American
5	Hispanic	Hispanic
6	African American	African American
7	Hispanic	Hispanic
⋮	⋮	⋮

Challenge #3: One category almost always occurs

If almost all of the observations fall into a single category the variable will offer little to no information in the analysis (too little variation). In some cases, models may fail to converge.



Challenge #4: Your outcome is categorical

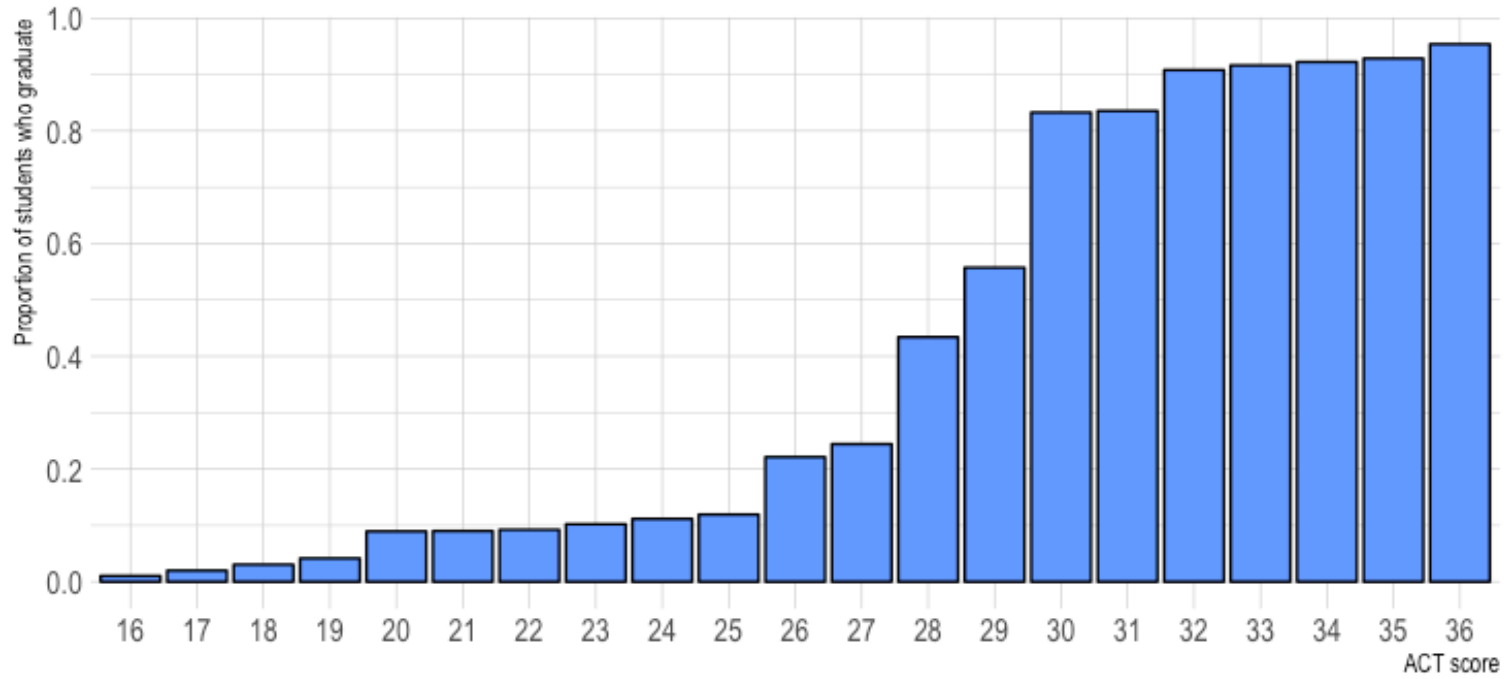
When your outcome is categorical, the linear models (e.g., regression, ANOVA, *t*-tests) are no longer appropriate for analyzing your data.



Imagine a researcher examining whether ACT score is predictive of whether or not students graduate college. In this analysis the outcome, graduation = Yes / No, is a dichotomous categorical variable.

A plot of the proportion of students who graduate by ACT score illustrates several problems with using methods that are meant for quantitative data:

- The curve that models the proportion of students who graduate is **S-shaped**; not linear.
- This is even more apparent if we extrapolate to really low or really high ACT scores; the proportion of students who graduate can never go below 0 or above 1 (they are bounds/asymptotes for our curve).
- If we are interested in inference, one of the assumptions of the linear model is *conditional normality*; proportions are not normally distributed—they are binomially distributed.



Potential Solution: Use methods that accommodate categorical outcomes.

- Bar charts
- Mosaic plots
- Biserial or Point-Biserial correlation coefficients
- Goodman and Kruskal's Lambda
- Chi-square tests of association/independence
- Tests of Proportion
- Generalized models (e.g., logistic regression)
- ROC analysis
- Survival models
- Classification trees



References and Resources

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